DBMS PROJECT REPORT

Phase 1

Student Skill–Project Matching System

DBMS

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Database Management Systems Project

1. Problem Statement

Students often have different technical skills, levels of proficiency, and areas of interest, while projects have their own skill requirements. When this information is kept separately, it becomes difficult to identify suitable students for a particular project and to keep track of applications and final project members.

The aim of this project is to build a structured database for managing student skills and project requirements. The system keeps student, department, skill, client, project, application, matching, and project membership information in related tables. It can then be used to find students whose skills are suitable for a project's requirements.

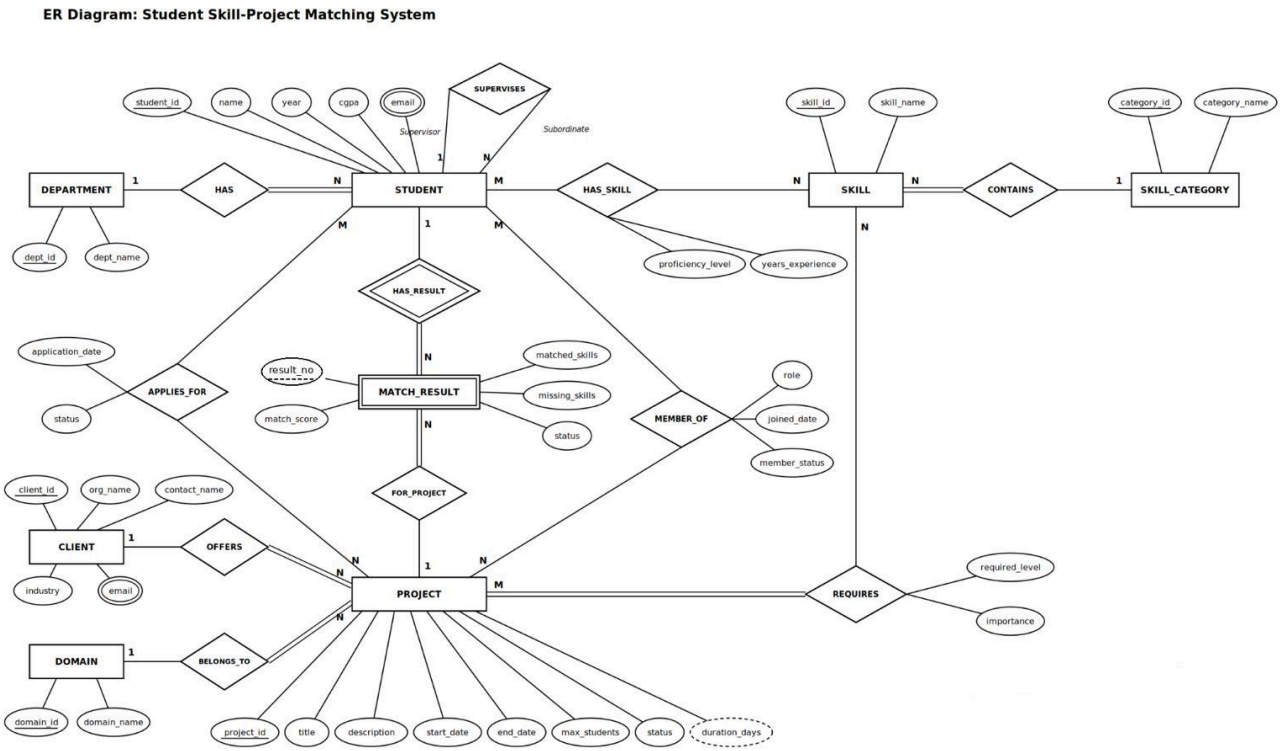
2. Brief Description of the Project

The Student Skill–Project Matching System is implemented as a relational database using MySQL. The database stores student details, departments, skill categories, skills, clients, domains, and projects. Students can have multiple skills, and projects can require multiple skills.

The Student–Skill relationship records proficiency level and years of experience. Similarly, the Project–Skill relationship records the required level and importance of a skill for a project. Applications record students applying for projects, Match_Result stores the result of matching a student with a project, and Project_Member records students who are selected for a project.

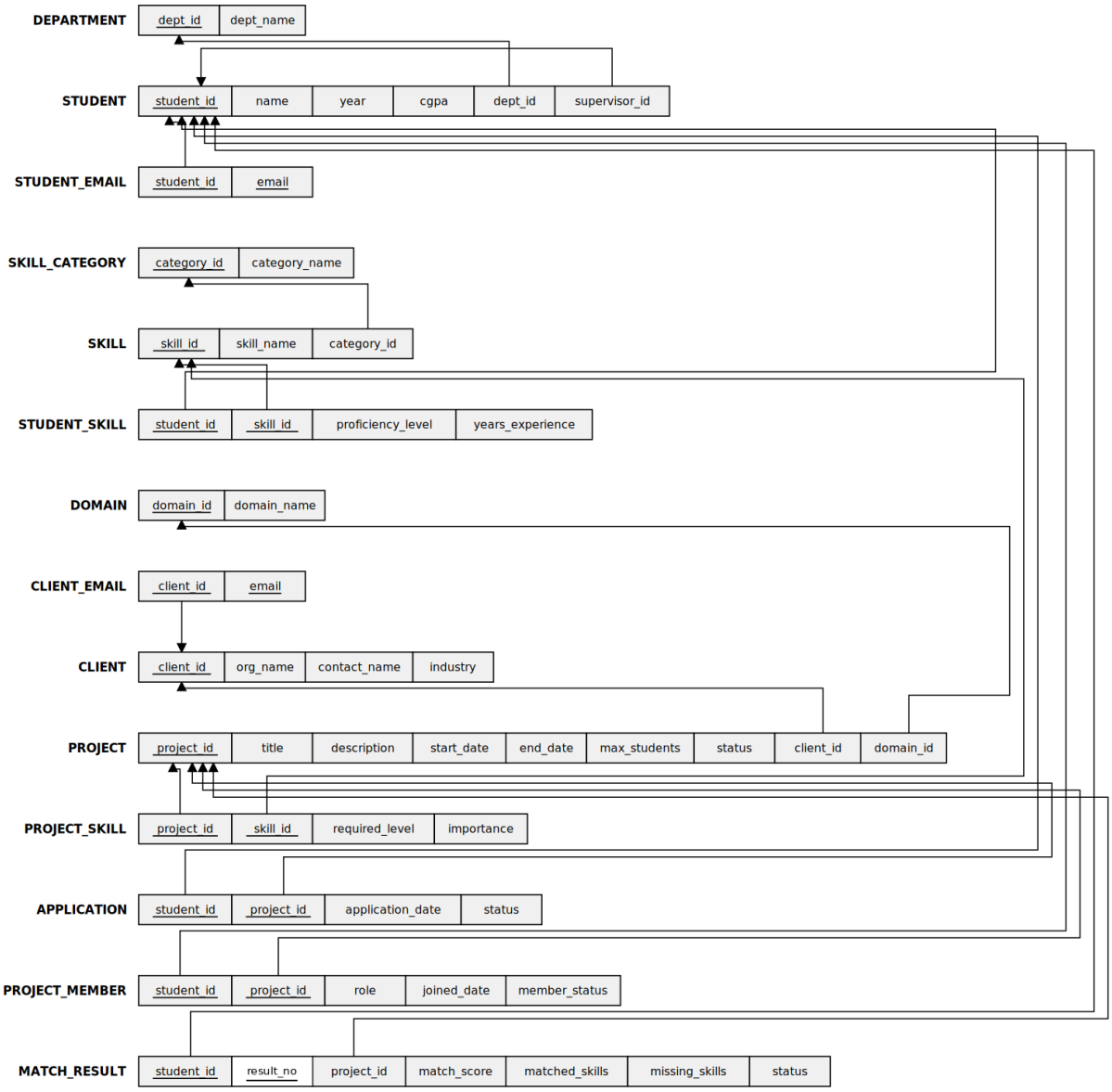
The design also demonstrates important ER-model concepts such as strong entities, a weak entity, an identifying relationship, multivalued attributes, a derived attribute, a recursive relationship, relationship attributes, primary keys, foreign keys, cardinality, and participation.

3. ER Diagram


ER Diagram: Student Skill-Project Matching System
student_id
name
year
cgpa
email
SUPERVISES
Supervisor
Subordinate
DEPARTMENT
HAS
STUDENT
HAS_SKILL
SKILL
CONTAINS
SKILL_CATEGORY
skill_id
skill_name
category_id
category_name
dept_id
dept_name
proficiency_level
years_experience
HAS_RESULT
application_date
APPLIES_FOR
status
result_no
MATCH_RESULT
matched_skills
missing_skills
match_score
status
MEMBER_OF
role
joined_date
member_status
client_id
org_name
contact_name
CLIENT
OFFERS
industry
email
FOR_PROJECT
PROJECT
REQUIRES
required_level
importance
DOMAIN
BELONGS_TO
domain_id
domain_name
project_id
title
description
start_date
end_date
max_students
status
duration_days

4. Relational Model

Relational Schema: Student Skill-Project Matching System



5. DDL Commands

```
CREATE DATABASE student_skill_project_matching;
USE student_skill_project_matching;

CREATE TABLE DEPARTMENT (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(100) NOT NULL UNIQUE
);
```

```
CREATE TABLE STUDENT (
    student_id INT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    year INT NOT NULL,
    cgpa DECIMAL(3,2),
    dept_id INT NOT NULL,
    supervisor_id INT,
    CONSTRAINT chk_student_year CHECK (year BETWEEN 1 AND 4),
    CONSTRAINT chk_student_cgpa CHECK (cgpa BETWEEN 0 AND 10),
    CONSTRAINT fk_student_department
        FOREIGN KEY (dept_id) REFERENCES DEPARTMENT(dept_id),
    CONSTRAINT fk_student_supervisor
        FOREIGN KEY (supervisor_id) REFERENCES STUDENT(student_id)
);

CREATE TABLE STUDENT_EMAIL (
    student_id INT,
    email VARCHAR(150),
    PRIMARY KEY (student_id, email),
    CONSTRAINT fk_student_email
        FOREIGN KEY (student_id) REFERENCES STUDENT(student_id)
        ON DELETE CASCADE
);

CREATE TABLE SKILL_CATEGORY (
    category_id INT PRIMARY KEY,
    category_name VARCHAR(100) NOT NULL UNIQUE
);

CREATE TABLE SKILL (
    skill_id INT PRIMARY KEY,
    skill_name VARCHAR(100) NOT NULL UNIQUE,
    category_id INT NOT NULL,
    CONSTRAINT fk_skill_category
        FOREIGN KEY (category_id) REFERENCES SKILL_CATEGORY(category_id)
);

CREATE TABLE STUDENT_SKILL (
    student_id INT,
    skill_id INT,
    proficiency_level VARCHAR(50) NOT NULL,
    years_experience DECIMAL(4,1),
    PRIMARY KEY (student_id, skill_id),
    CONSTRAINT fk_student_skill_student
        FOREIGN KEY (student_id) REFERENCES STUDENT(student_id)
        ON DELETE CASCADE,
    CONSTRAINT fk_student_skill_skill
        FOREIGN KEY (skill_id) REFERENCES SKILL(skill_id)
        ON DELETE CASCADE,
    CONSTRAINT chk_student_skill_experience
        CHECK (years_experience >= 0)
);

CREATE TABLE CLIENT (
    client_id INT PRIMARY KEY,
    org_name VARCHAR(150) NOT NULL,
    contact_name VARCHAR(100) NOT NULL,
    industry VARCHAR(100)
```

```
);

CREATE TABLE CLIENT_EMAIL (
    client_id INT,
    email VARCHAR(150),
    PRIMARY KEY (client_id, email),
    CONSTRAINT fk_client_email
        FOREIGN KEY (client_id) REFERENCES CLIENT(client_id)
        ON DELETE CASCADE
);

CREATE TABLE DOMAIN (
    domain_id INT PRIMARY KEY,
    domain_name VARCHAR(100) NOT NULL UNIQUE
);

CREATE TABLE PROJECT (
    project_id INT PRIMARY KEY,
    title VARCHAR(200) NOT NULL,
    description TEXT,
    start_date DATE,
    end_date DATE,
    max_students INT,
    status VARCHAR(50) NOT NULL,
    client_id INT NOT NULL,
    domain_id INT NOT NULL,
    CONSTRAINT fk_project_client
        FOREIGN KEY (client_id) REFERENCES CLIENT(client_id),
    CONSTRAINT fk_project_domain
        FOREIGN KEY (domain_id) REFERENCES DOMAIN(domain_id),
    CONSTRAINT chk_project_dates
        CHECK (end_date IS NULL OR start_date IS NULL OR end_date >= start_date),
    CONSTRAINT chk_project_max_students
        CHECK (max_students > 0)
);

CREATE TABLE PROJECT_SKILL (
    project_id INT,
    skill_id INT,
    required_level VARCHAR(50) NOT NULL,
    importance VARCHAR(50),
    PRIMARY KEY (project_id, skill_id),
    CONSTRAINT fk_project_skill_project
        FOREIGN KEY (project_id) REFERENCES PROJECT(project_id)
        ON DELETE CASCADE,
    CONSTRAINT fk_project_skill_skill
        FOREIGN KEY (skill_id) REFERENCES SKILL(skill_id)
        ON DELETE CASCADE
);

CREATE TABLE APPLICATION (
    student_id INT,
    project_id INT,
    application_date DATE NOT NULL,
    status VARCHAR(50) NOT NULL,
    PRIMARY KEY (student_id, project_id),
    CONSTRAINT fk_application_student
        FOREIGN KEY (student_id) REFERENCES STUDENT(student_id)
        ON DELETE CASCADE,
```

```
    CONSTRAINT fk_application_project
        FOREIGN KEY (project_id) REFERENCES PROJECT(project_id)
        ON DELETE CASCADE
);

CREATE TABLE PROJECT_MEMBER (
    student_id INT,
    project_id INT,
    role VARCHAR(100) NOT NULL,
    joined_date DATE,
    member_status VARCHAR(50) NOT NULL,
    PRIMARY KEY (student_id, project_id),
    CONSTRAINT fk_project_member_student
        FOREIGN KEY (student_id) REFERENCES STUDENT(student_id)
        ON DELETE CASCADE,
    CONSTRAINT fk_project_member_project
        FOREIGN KEY (project_id) REFERENCES PROJECT(project_id)
        ON DELETE CASCADE
);

CREATE TABLE MATCH_RESULT (
    student_id INT,
    result_no INT,
    project_id INT NOT NULL,
    match_score DECIMAL(5,2),
    matched_skills TEXT,
    missing_skills TEXT,
    status VARCHAR(50) NOT NULL,
    PRIMARY KEY (student_id, result_no),
    CONSTRAINT fk_match_result_student
        FOREIGN KEY (student_id) REFERENCES STUDENT(student_id)
        ON DELETE CASCADE,
    CONSTRAINT fk_match_result_project
        FOREIGN KEY (project_id) REFERENCES PROJECT(project_id)
        ON DELETE CASCADE,
    CONSTRAINT chk_match_score
        CHECK (match_score BETWEEN 0 AND 100),
    CONSTRAINT chk_result_no
        CHECK (result_no > 0)
);
```

```
mysql> SHOW TABLES;
+-----------------------------------------+
| Tables_in_student_skill_project_matching |
+-----------------------------------------+
| APPLICATION                             |
| CLIENT                                  |
| CLIENT_EMAIL                            |
| DEPARTMENT                              |
| DOMAIN                                  |
| MATCH_RESULT                            |
| PROJECT                                 |
| PROJECT_MEMBER                          |
| PROJECT_SKILL                           |
| SKILL                                   |
| SKILL_CATEGORY                          |
| STUDENT                                 |
| STUDENT_EMAIL                           |
| STUDENT_SKILL                           |
+-----------------------------------------+
14 rows in set (0.00 sec)
```

```
mysql> DESCRIBE STUDENT;
+---------------+--------------+------+-----+---------+-------+
| Field         | Type         | Null | Key | Default | Extra |
+---------------+--------------+------+-----+---------+-------+
| student_id    | int          | NO   | PRI | NULL    |       |
| name          | varchar(100) | NO   |     | NULL    |       |
| year          | int          | NO   |     | NULL    |       |
| cgpa          | decimal(3,2) | YES  |     | NULL    |       |
| dept_id       | int          | NO   | MUL | NULL    |       |
| supervisor_id | int          | YES  | MUL | NULL    |       |
+---------------+--------------+------+-----+---------+-------+
6 rows in set (0.01 sec)
```

```
mysql> DESCRIBE MATCH_RESULT;
+----------------+--------------+------+-----+---------+-------+
| Field          | Type         | Null | Key | Default | Extra |
+----------------+--------------+------+-----+---------+-------+
| student_id     | int          | NO   | PRI | NULL    |       |
| result_no      | int          | NO   | PRI | NULL    |       |
| project_id     | int          | NO   | MUL | NULL    |       |
| match_score    | decimal(5,2) | YES  |     | NULL    |       |
| matched_skills | text         | YES  |     | NULL    |       |
| missing_skills | text         | YES  |     | NULL    |       |
| status         | varchar(50)  | NO   |     | NULL    |       |
+----------------+--------------+------+-----+---------+-------+
7 rows in set (0.00 sec)
```



```
mysql> SHOW CREATE TABLE MATCH_RESULT;
+--------------+-----------------------
```

```
(3, 'Electronics and Communication');

INSERT INTO STUDENT
(student_id, name, year, cgpa, dept_id, supervisor_id)
VALUES
(101, 'Rahul', 3, 8.70, 1, NULL),
(102, 'Ananya', 3, 9.10, 1, 101),
(103, 'Vikram', 2, 8.40, 2, 101),
(104, 'Sneha', 4, 9.30, 1, 101),
(105, 'Arjun', 2, 7.90, 3, NULL);

INSERT INTO STUDENT_EMAIL
(student_id, email)
VALUES
(101, 'rahul@gmail.com'),
(101, 'rahul.college@gmail.com'),
(102, 'ananya@gmail.com'),
(103, 'vikram@gmail.com'),
(104, 'sneha@gmail.com'),
(105, 'arjun@gmail.com');

INSERT INTO SKILL_CATEGORY
(category_id, category_name)
VALUES
(1, 'Programming'),
(2, 'Database'),
(3, 'Web Development'),
(4, 'Data Science');

INSERT INTO SKILL
(skill_id, skill_name, category_id)
VALUES
(201, 'Python', 1),
(202, 'Java', 1),
(203, 'SQL', 2),
(204, 'MySQL', 2),
(205, 'HTML', 3),
(206, 'CSS', 3),
(207, 'Machine Learning', 4);

INSERT INTO STUDENT_SKILL
(student_id, skill_id, proficiency_level, years_experience)
VALUES
(101, 201, 'Advanced', 2.0),
(101, 203, 'Advanced', 2.5),
(101, 204, 'Intermediate', 1.5),
(102, 201, 'Advanced', 2.0),
(102, 207, 'Advanced', 1.5),
(103, 202, 'Intermediate', 1.0),
(103, 203, 'Advanced', 2.0),
(104, 201, 'Expert', 3.0),
(104, 203, 'Advanced', 3.0),
(104, 207, 'Advanced', 2.0),
(105, 205, 'Advanced', 1.5),
(105, 206, 'Intermediate', 1.0);

INSERT INTO CLIENT
(client_id, org_name, contact_name, industry)
VALUES
```

```
(301, 'TechNova Solutions', 'Kiran Kumar', 'Software'),
(302, 'DataWorks Pvt Ltd', 'Meera Rao', 'Data Analytics'),
(303, 'WebSphere Technologies', 'Rohit Sharma', 'Web Development');

INSERT INTO CLIENT_EMAIL
(client_id, email)
VALUES
(301, 'contact@technova.com'),
(301, 'hr@technova.com'),
(302, 'contact@dataworks.com'),
(303, 'projects@websphere.com');

INSERT INTO DOMAIN
(domain_id, domain_name)
VALUES
(401, 'Artificial Intelligence'),
(402, 'Web Development'),
(403, 'Database Systems'),
(404, 'Data Analytics');

INSERT INTO PROJECT
(project_id, title, description, start_date, end_date,
 max_students, status, client_id, domain_id)
VALUES
(501, 'AI Student Assistant',
 'Development of an AI based student assistant.',
 '2026-01-10', '2026-04-10',
 4, 'Open', 301, 401),
(502, 'College Website',
 'Development of a responsive college website.',
 '2026-02-01', '2026-05-01',
 5, 'Open', 303, 402),
(503, 'Student Database System',
 'Design and implementation of a student database.',
 '2026-01-15', '2026-04-15',
 3, 'Open', 301, 403),
(504, 'Data Analytics Dashboard',
 'Development of a dashboard for student analytics.',
 '2026-03-01', '2026-06-01',
 4, 'Open', 302, 404);

INSERT INTO PROJECT_SKILL
(project_id, skill_id, required_level, importance)
VALUES
(501, 201, 'Advanced', 'High'),
(501, 207, 'Advanced', 'High'),
(501, 203, 'Intermediate', 'Medium'),
(502, 205, 'Advanced', 'High'),
(502, 206, 'Intermediate', 'Medium'),
(503, 203, 'Advanced', 'High'),
(503, 204, 'Advanced', 'High'),
(503, 202, 'Intermediate', 'Medium'),
(504, 201, 'Advanced', 'High'),
(504, 203, 'Intermediate', 'Medium'),
(504, 207, 'Advanced', 'High');

INSERT INTO APPLICATION
(student_id, project_id, application_date, status)
VALUES
```

```
(101, 501, '2026-01-05', 'Applied'),
(101, 503, '2026-01-06', 'Shortlisted'),
(102, 501, '2026-01-07', 'Applied'),
(102, 504, '2026-02-10', 'Applied'),
(103, 503, '2026-01-10', 'Applied'),
(104, 501, '2026-01-08', 'Selected'),
(104, 504, '2026-02-12', 'Shortlisted'),
(105, 502, '2026-01-20', 'Applied');

INSERT INTO PROJECT_MEMBER
(student_id, project_id, role, joined_date, member_status)
VALUES
(104, 501, 'AI Developer', '2026-01-10', 'Active'),
(101, 503, 'Database Developer', '2026-01-15', 'Active'),
(105, 502, 'Frontend Developer', '2026-02-01', 'Active');

INSERT INTO MATCH_RESULT
(student_id, result_no, project_id, match_score,
 matched_skills, missing_skills, status)
VALUES
(101, 1, 501, 78.50,
 'Python, SQL', 'Machine Learning', 'Matched'),
(101, 2, 503, 92.00,
 'SQL, MySQL', 'Java', 'Highly Matched'),
(102, 1, 501, 95.00,
 'Python, Machine Learning', 'SQL', 'Highly Matched'),
(102, 2, 504, 88.00,
 'Python, Machine Learning', 'SQL', 'Matched'),
(103, 1, 503, 72.00,
 'SQL', 'MySQL, Java', 'Partially Matched'),
(104, 1, 501, 98.00,
 'Python, SQL, Machine Learning', '', 'Highly Matched'),
(104, 2, 504, 96.00,
 'Python, SQL, Machine Learning', '', 'Highly Matched'),
(105, 1, 502, 91.00,
 'HTML, CSS', '', 'Highly Matched');
```

6.2 Data Retrieval

```
SELECT *

FROM STUDENT;



SELECT

    s.student_id,

    s.name,

    d.dept_name

FROM STUDENT s

JOIN DEPARTMENT d

ON s.dept_id = d.dept_id;
```

```
SELECT
    s.name,
    sk.skill_name,
    ss.proficiency_level,
    ss.years_experience
FROM STUDENT_SKILL ss
JOIN STUDENT s
ON ss.student_id = s.student_id
JOIN SKILL sk
ON ss.skill_id = sk.skill_id;

SELECT
    p.title,
    sk.skill_name,
    ps.required_level,
    ps.importance
FROM PROJECT_SKILL ps
JOIN PROJECT p
ON ps.project_id = p.project_id
JOIN SKILL sk
ON ps.skill_id = sk.skill_id;

SELECT
    s.name,
    p.title,
    a.application_date,
    a.status
FROM APPLICATION a
JOIN STUDENT s
ON a.student_id = s.student_id
JOIN PROJECT p
```

```
ON a.project_id = p.project_id;


SELECT
    s.name,
    p.title,
    mr.result_no,
    mr.match_score,
    mr.matched_skills,
    mr.missing_skills,
    mr.status
FROM MATCH_RESULT mr
JOIN STUDENT s
ON mr.student_id = s.student_id
JOIN PROJECT p
ON mr.project_id = p.project_id
ORDER BY mr.match_score DESC;


SELECT
    s.name,
    p.title,
    pm.role,
    pm.joined_date,
    pm.member_status
FROM PROJECT_MEMBER pm
JOIN STUDENT s
ON pm.student_id = s.student_id
JOIN PROJECT p
ON pm.project_id = p.project_id;
```

6.3 Data Modification

```
UPDATE APPLICATION
SET status = 'Selected'
```

```
WHERE student_id = 104
AND project_id = 501;

DELETE FROM APPLICATION
WHERE student_id = 105
AND project_id = 502;
```

```
mysql> INSERT INTO DEPARTMENT
    -> (dept_id, dept_name)
    -> VALUES
    -> (1, 'Computer Science'),
    -> (2, 'Information Science'),
    -> (3, 'Electronics and Communication');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM DEPARTMENT;
+---------+-------------------------------+
| dept_id | dept_name                     |
+---------+-------------------------------+
|       1 | Computer Science              |
|       3 | Electronics and Communication |
|       2 | Information Science           |
+---------+-------------------------------+
3 rows in set (0.01 sec)
```

```
mysql> INSERT INTO STUDENT
    -> (student_id, name, year, cgpa, dept_id, supervisor_id)
    -> VALUES
    -> (101, 'Rahul', 3, 8.70, 1, NULL),
    -> (102, 'Ananya', 3, 9.10, 1, 101),
    -> (103, 'Vikram', 2, 8.40, 2, 101),
    -> (104, 'Sneha', 4, 9.30, 1, 101),
    -> (105, 'Arjun', 2, 7.90, 3, NULL);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM STUDENT;
+------------+--------+------+------+---------+---------------+
| student_id | name   | year | cgpa | dept_id | supervisor_id |
+------------+--------+------+------+---------+---------------+
|        101 | Rahul  |    3 | 8.70 |       1 |          NULL |
|        102 | Ananya |    3 | 9.10 |       1 |           101 |
|        103 | Vikram |    2 | 8.40 |       2 |           101 |
|        104 | Sneha  |    4 | 9.30 |       1 |           101 |
|        105 | Arjun  |    2 | 7.90 |       3 |          NULL |
+------------+--------+------+------+---------+---------------+
5 rows in set (0.00 sec)
```

```
mysql> INSERT INTO STUDENT_EMAIL
    -> (student_id, email)
    -> VALUES
    -> (101, 'rahul@gmail.com'),
    -> (101, 'rahul.college@gmail.com'),
    -> (102, 'ananya@gmail.com'),
    -> (103, 'vikram@gmail.com'),
    -> (104, 'sneha@gmail.com'),
    -> (105, 'arjun@gmail.com');
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM STUDENT_EMAIL;
+------------+-------------------------+
| student_id | email                   |
+------------+-------------------------+
|        101 | rahul.college@gmail.com |
|        101 | rahul@gmail.com         |
|        102 | ananya@gmail.com        |
|        103 | vikram@gmail.com        |
|        104 | sneha@gmail.com         |
|        105 | arjun@gmail.com         |
+------------+-------------------------+
6 rows in set (0.00 sec)
```

```
mysql> INSERT INTO SKILL_CATEGORY
    -> (category_id, category_name)
    -> VALUES
    -> (1, 'Programming'),
    -> (2, 'Database'),
    -> (3, 'Web Development'),
    -> (4, 'Data Science');
Query OK, 4 rows affected (0.00 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM SKILL_CATEGORY;
+-------------+-----------------+
| category_id | category_name   |
+-------------+-----------------+
|           4 | Data Science    |
|           2 | Database        |
|           1 | Programming     |
|           3 | Web Development |
+-------------+-----------------+
4 rows in set (0.00 sec)
```

```
mysql> INSERT INTO SKILL
    -> (skill_id, skill_name, category_id)
    -> VALUES
    -> (201, 'Python', 1),
    -> (202, 'Java', 1),
    -> (203, 'SQL', 2),
    -> (204, 'MySQL', 2),
    -> (205, 'HTML', 3),
    -> (206, 'CSS', 3),
    -> (207, 'Machine Learning', 4);
Query OK, 7 rows affected (0.00 sec)
Records: 7  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM SKILL;
+----------+------------------+-------------+
| skill_id | skill_name       | category_id |
+----------+------------------+-------------+
|      201 | Python           |           1 |
|      202 | Java             |           1 |
|      203 | SQL              |           2 |
|      204 | MySQL            |           2 |
|      205 | HTML             |           3 |
|      206 | CSS              |           3 |
|      207 | Machine Learning |           4 |
+----------+------------------+-------------+
7 rows in set (0.00 sec)
```

```
mysql> INSERT INTO STUDENT_SKILL
    -> (student_id, skill_id, proficiency_level, years_experience)
    -> VALUES
    -> (101, 201, 'Advanced', 2.0),
    -> (101, 203, 'Advanced', 2.5),
    -> (101, 204, 'Intermediate', 1.5),
    -> (102, 201, 'Advanced', 2.0),
    -> (102, 207, 'Advanced', 1.5),
    -> (103, 202, 'Intermediate', 1.0),
    -> (103, 203, 'Advanced', 2.0),
    -> (104, 201, 'Expert', 3.0),
    -> (104, 203, 'Advanced', 3.0),
    -> (104, 207, 'Advanced', 2.0),
    -> (105, 205, 'Advanced', 1.5),
    -> (105, 206, 'Intermediate', 1.0);
Query OK, 12 rows affected (0.01 sec)
Records: 12  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM STUDENT_SKILL;
+------------+----------+-------------------+-------------------+
| student_id | skill_id | proficiency_level | years_experience |
+------------+----------+-------------------+-------------------+
|        101 |      201 | Advanced          |               2.0 |
|        101 |      203 | Advanced          |               2.5 |
|        101 |      204 | Intermediate      |               1.5 |
|        102 |      201 | Advanced          |               2.0 |
|        102 |      207 | Advanced          |               1.5 |
|        103 |      202 | Intermediate      |               1.0 |
|        103 |      203 | Advanced          |               2.0 |
|        104 |      201 | Expert            |               3.0 |
|        104 |      203 | Advanced          |               3.0 |
|        104 |      207 | Advanced          |               2.0 |
|        105 |      205 | Advanced          |               1.5 |
|        105 |      206 | Intermediate      |               1.0 |
+------------+----------+-------------------+-------------------+
12 rows in set (0.00 sec)

mysql> SELECT
    ->     s.name,
    ->     sk.skill_name,
    ->     ss.proficiency_level,
    ->     ss.years_experience
    -> FROM STUDENT_SKILL ss
    -> JOIN STUDENT s
    ->     ON ss.student_id = s.student_id
    -> JOIN SKILL sk
    ->     ON ss.skill_id = sk.skill_id
    -> ORDER BY s.student_id;
+--------+------------------+-------------------+-------------------+
| name   | skill_name       | proficiency_level | years_experience |
+--------+------------------+-------------------+-------------------+
| Rahul  | Python           | Advanced          |               2.0 |
| Rahul  | SQL              | Advanced          |               2.5 |
| Rahul  | MySQL            | Intermediate      |               1.5 |
| Ananya | Python           | Advanced          |               2.0 |
| Ananya | Machine Learning | Advanced          |               1.5 |
| Vikram | Java             | Intermediate      |               1.0 |
| Vikram | SQL              | Advanced          |               2.0 |
| Sneha  | Python           | Expert            |               3.0 |
| Sneha  | SQL              | Advanced          |               3.0 |
| Sneha  | Machine Learning | Advanced          |               2.0 |
| Arjun  | HTML             | Advanced          |               1.5 |
| Arjun  | CSS              | Intermediate      |               1.0 |
+--------+------------------+-------------------+-------------------+
12 rows in set (0.00 sec)
```

```
mysql> INSERT INTO PROJECT_SKILL
    -> (project_id, skill_id, required_level, importance)
    -> VALUES
    -> (501, 201, 'Advanced', 'High'),
    -> (501, 207, 'Advanced', 'High'),
    -> (501, 203, 'Intermediate', 'Medium'),
    ->
    -> (502, 205, 'Advanced', 'High'),
    -> (502, 206, 'Intermediate', 'Medium'),
    ->
    -> (503, 203, 'Advanced', 'High'),
    -> (503, 204, 'Advanced', 'High'),
    -> (503, 202, 'Intermediate', 'Medium'),
    ->
    -> (504, 201, 'Advanced', 'High'),
    -> (504, 203, 'Intermediate', 'Medium'),
    -> (504, 207, 'Advanced', 'High');
Query OK, 11 rows affected (0.01 sec)
Records: 11  Duplicates: 0  Warnings: 0

mysql> SELECT
    ->     p.title,
    ->     sk.skill_name,
    ->     ps.required_level,
    ->     ps.importance
    -> FROM PROJECT_SKILL ps
    -> JOIN PROJECT p
    ->     ON ps.project_id = p.project_id
    -> JOIN SKILL sk
    ->     ON ps.skill_id = sk.skill_id
    -> ORDER BY p.project_id;
+--------------------------+------------------+----------------+------------+
| title                    | skill_name       | required_level | importance |
+--------------------------+------------------+----------------+------------+
| AI Student Assistant     | Python           | Advanced       | High       |
| AI Student Assistant     | SQL              | Intermediate   | Medium     |
| AI Student Assistant     | Machine Learning | Advanced       | High       |
| College Website          | HTML             | Advanced       | High       |
| College Website          | CSS              | Intermediate   | Medium     |
| Student Database System  | Java             | Intermediate   | Medium     |
| Student Database System  | SQL              | Advanced       | High       |
| Student Database System  | MySQL            | Advanced       | High       |
| Data Analytics Dashboard | Python           | Advanced       | High       |
| Data Analytics Dashboard | SQL              | Intermediate   | Medium     |
| Data Analytics Dashboard | Machine Learning | Advanced       | High       |
+--------------------------+------------------+----------------+------------+
11 rows in set (0.00 sec)
```

```
mysql> INSERT INTO APPLICATION
    -> (student_id, project_id, application_date, status)
    -> VALUES
    -> (101, 501, '2026-01-05', 'Applied'),
    -> (101, 503, '2026-01-06', 'Shortlisted'),
    -> (102, 501, '2026-01-07', 'Applied'),
    -> (102, 504, '2026-02-10', 'Applied'),
    -> (103, 503, '2026-01-10', 'Applied'),
    -> (104, 501, '2026-01-08', 'Selected'),
    -> (104, 504, '2026-02-12', 'Shortlisted'),
    -> (105, 502, '2026-01-20', 'Applied');
Query OK, 8 rows affected (0.01 sec)
Records: 8  Duplicates: 0  Warnings: 0

mysql> SELECT
    ->     s.name,
    ->     p.title,
    ->     a.application_date,
    ->     a.status
    -> FROM APPLICATION a
    -> JOIN STUDENT s
    ->     ON a.student_id = s.student_id
    -> JOIN PROJECT p
    ->     ON a.project_id = p.project_id
    -> ORDER BY a.application_date;
+--------+--------------------------+------------------+-------------+
| name   | title                    | application_date | status      |
+--------+--------------------------+------------------+-------------+
| Rahul  | AI Student Assistant     | 2026-01-05       | Applied     |
| Rahul  | Student Database System  | 2026-01-06       | Shortlisted |
| Ananya | AI Student Assistant     | 2026-01-07       | Applied     |
| Sneha  | AI Student Assistant     | 2026-01-08       | Selected    |
| Vikram | Student Database System  | 2026-01-10       | Applied     |
| Arjun  | College Website          | 2026-01-20       | Applied     |
| Ananya | Data Analytics Dashboard | 2026-02-10       | Applied     |
| Sneha  | Data Analytics Dashboard | 2026-02-12       | Shortlisted |
+--------+--------------------------+------------------+-------------+
8 rows in set (0.00 sec)
```

```
mysql> INSERT INTO PROJECT_MEMBER
    -> (student_id, project_id, role, joined_date, member_status)
    -> VALUES
    -> (104, 501, 'AI Developer', '2026-01-10', 'Active'),
    -> (101, 503, 'Database Developer', '2026-01-15', 'Active'),
    -> (105, 502, 'Frontend Developer', '2026-02-01', 'Active');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT
    ->     s.name,
    ->     p.title,
    ->     pm.role,
    ->     pm.joined_date,
    ->     pm.member_status
    -> FROM PROJECT_MEMBER pm
    -> JOIN STUDENT s
    ->     ON pm.student_id = s.student_id
    -> JOIN PROJECT p
    ->     ON pm.project_id = p.project_id;
+-------+-------------------------+--------------------+-------------+---------------+
| name  | title                   | role               | joined_date | member_status |
+-------+-------------------------+--------------------+-------------+---------------+
| Rahul | Student Database System | Database Developer | 2026-01-15  | Active        |
| Sneha | AI Student Assistant    | AI Developer       | 2026-01-10  | Active        |
| Arjun | College Website         | Frontend Developer | 2026-02-01  | Active        |
+-------+-------------------------+--------------------+-------------+---------------+
3 rows in set (0.00 sec)
```

```
mysql> INSERT INTO MATCH_RESULT
    -> (student_id, result_no, project_id, match_score,
    ->  matched_skills, missing_skills, status)
    -> VALUES
    -> (101, 1, 501, 78.50,
    ->  'Python, SQL',
    ->  'Machine Learning',
    ->  'Matched'),
    ->
    -> (101, 2, 503, 92.00,
    ->  'SQL, MySQL',
    ->  'Java',
    ->  'Highly Matched'),
    ->
    -> (102, 1, 501, 95.00,
    ->  'Python, Machine Learning',
    ->  'SQL',
    ->  'Highly Matched'),
    ->
    -> (102, 2, 504, 88.00,
    ->  'Python, Machine Learning',
    ->  'SQL',
    ->  'Matched'),
    ->
    -> (103, 1, 503, 72.00,
    ->  'SQL',
    ->  'MySQL, Java',
    ->  'Partially Matched'),
    ->
    -> (104, 1, 501, 98.00,
    ->  'Python, SQL, Machine Learning',
    ->  '',
    ->  'Highly Matched'),
    ->
    -> (104, 2, 504, 96.00,
    ->  'Python, SQL, Machine Learning',
    ->  '',
    ->  'Highly Matched'),
    ->
    -> (105, 1, 502, 91.00,
    ->  'HTML, CSS',
    ->  '',
    ->  'Highly Matched');
Query OK, 8 rows affected (0.00 sec)
Records: 8  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT
    ->     s.name,
    ->     p.title,
    ->     mr.result_no,
    ->     mr.match_score,
    ->     mr.matched_skills,
    ->     mr.missing_skills,
    ->     mr.status
    -> FROM MATCH_RESULT mr
    -> JOIN STUDENT s
    ->     ON mr.student_id = s.student_id
    -> JOIN PROJECT p
    ->     ON mr.project_id = p.project_id
    -> ORDER BY mr.match_score DESC;
+--------+--------------------------+-----------+-------------+------------------------------+-----------------
-+-------------------+
| name   | title                    | result_no | match_score | matched_skills               | missing_skills
 | status            |
+--------+--------------------------+-----------+-------------+------------------------------+-----------------
-+-------------------+
| Sneha  | AI Student Assistant     |         1 |       98.00 | Python, SQL, Machine Learning |
 | Highly Matched    |
| Sneha  | Data Analytics Dashboard |         2 |       96.00 | Python, SQL, Machine Learning |
 | Highly Matched    |
| Ananya | AI Student Assistant     |         1 |       95.00 | Python, Machine Learning      | SQL
 | Highly Matched    |
| Rahul  | Student Database System  |         2 |       92.00 | SQL, MySQL                    | Java
 | Highly Matched    |
| Arjun  | College Website          |         1 |       91.00 | HTML, CSS                     |
 | Highly Matched    |
| Ananya | Data Analytics Dashboard |         2 |       88.00 | Python, Machine Learning      | SQL
 | Matched           |
| Rahul  | AI Student Assistant     |         1 |       78.50 | Python, SQL                   | Machine Learning
 | Matched           |
| Vikram | Student Database System  |         1 |       72.00 | SQL                           | MySQL, Java
 | Partially Matched |
+--------+--------------------------+-----------+-------------+------------------------------+-----------------
-+-------------------+
8 rows in set (0.00 sec)

mysql>
```

```
mysql> UPDATE APPLICATION
    -> SET status = 'Selected'
    -> WHERE student_id = 104
    -> AND project_id = 501;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0

mysql> SELECT *
    -> FROM APPLICATION
    -> WHERE student_id = 104
    -> AND project_id = 501;
+------------+------------+------------------+-----------+
| student_id | project_id | application_date | status    |
+------------+------------+------------------+-----------+
|        104 |        501 | 2026-01-08       | Selected  |
+------------+------------+------------------+-----------+
1 row in set (0.00 sec)
```

```
mysql> SELECT *
    -> FROM APPLICATION
    -> WHERE student_id = 105
    -> AND project_id = 502;
+------------+------------+------------------+---------+
| student_id | project_id | application_date | status  |
+------------+------------+------------------+---------+
|        105 |        502 | 2026-01-20       | Applied |
+------------+------------+------------------+---------+
1 row in set (0.00 sec)

mysql> DELETE FROM APPLICATION
    -> WHERE student_id = 105
    -> AND project_id = 502;
Query OK, 1 row affected (0.00 sec)

mysql> SELECT *
    -> FROM APPLICATION
    -> WHERE student_id = 105
    -> AND project_id = 502;
Empty set (0.00 sec)

mysql>
```

7. Main Relationships

• DEPARTMENT 1:N STUDENT — one department can have many students.

• STUDENT 1:N STUDENT — recursive SUPERVISES relationship using Supervisor and Subordinate roles.

• SKILL_CATEGORY 1:N SKILL — one category contains many skills.

• STUDENT M:N SKILL — represented through STUDENT_SKILL; proficiency_level and years_experience describe the relationship.

• CLIENT 1:N PROJECT — one client can offer many projects.

• DOMAIN 1:N PROJECT — one domain can contain many projects.

• PROJECT M:N SKILL — represented through PROJECT_SKILL; required_level and importance describe the relationship.

• STUDENT M:N PROJECT — represented through APPLICATION.

• STUDENT 1:N MATCH_RESULT — MATCH_RESULT is modeled as a weak entity dependent on STUDENT and identified using result_no.

• MATCH_RESULT N:1 PROJECT — each match result is associated with a project.

• STUDENT M:N PROJECT — represented through PROJECT_MEMBER for selected project members.

8. Conclusion

The Student Skill–Project Matching System provides a structured way to store student skills and project requirements and to keep track of applications, matching results, and project membership. The ER model captures the relationships between the different parts of the system, while the relational model translates those relationships into tables with primary and foreign keys.

The database also demonstrates important DBMS concepts such as DDL, DML, integrity constraints, many-to-many relationship mapping, multivalued attributes, a weak entity, recursive relationships, and relationship attributes. These concepts make the database suitable for demonstrating the fundamentals of relational database design.